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ORIGINAL ARTICLE

Isolation, structure analysis and expression characterization of the Hexokinase gene family in *Sorghum bicolor*

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Abstract

Hexokinases (HXK) not only facilitate carbohydrate metabolism but also play important roles in sugar sensing in higher plants. *HXK* gene families have been extensively discussed in many plant species; however, comprehensive information regarding *HXKs* in sorghum remains unclear. To address this gap, we identified 7 putative sorghum *HXKs* (*SbHXK1* to *SbHXK7*), and the features of their conserved domains, gene structure, evolutionary tree, and cis-acting elements were systematically characterized to reveal the evolutionary conservation between different plant species. Based on expression profiling, we found that different expression patterns of *SbHXKs* were associated with different physiological processes, including abiotic stresses. Further qRT-PCR verification under salt and sucrose treatment confirmed that *SbHXK2*, *SbHXK4*, and *SbHXK5* may play very important roles under high osmotic pressure. Notably, SbHXKs are predominantly localized in the cytoplasm, in contrast to some rice and *Arabidopsis* HXKs, which are localized in chloroplasts or mitochondria, suggesting divergent roles for SbHXKs. In summary, our study provides a theoretical foundation for understanding the HXK gene family and offers fundamental insights of *SbHXKs* in sorghum.

Keywords Hexokinases · *Sorghum bicolor* (L.) Moench · Gene family · Abiotic stress · Expression analysis

Abbreviations

HXK	Hexokinase
qRT-PCR	Quantitative real-time-polymerase chain reaction
CDS	Coding sequence
MW	Molecular weight
PI	Isoelectric point
ABA	Abscisic acid
MYC	Myelo Cytomatosis
ABRE	ABA response element
STRE	Stress-responsive element
AS-1	Activation sequence-1

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Introduction

Sorghum (*Sorghum bicolor* (L.) Moench) is a highly efficient C4 crop, serving not only as a source of food and feed, but also as a crucial bioenergy resource (Wang et al. 2013; Taylor et al. 2014). Consequently, enhancing both yield and quality remains a primary breeding objective for sorghum. Carbohydrate metabolism plays a central role in achieving this goal (Schittenhelm et al. 2004). Carbohydrates, derived from photosynthesis, fuel various metabolic processes essential for synthesizing fats, proteins, and nucleic acids, thereby supporting plant growth. In higher plants, sugars serve as the primary source of carbon and energy, primarily synthesized and stored in photosynthetic tissues (Chen et al. 2015; Wang et al. 2018).

In higher plants, glucose and other hexoses serve as the universal carbon source and play a crucial role as sugar signaling molecules in various biological processes. Hexokinase (HXK) is the first rate-limiting enzyme involved in the glycolysis pathway, phosphorylating and catalyzing hexoses form hexose 6-phosphates (Li et al. 2017). Therefore, HXKs are considered as the key enzymes in regulating carbohydrate metabolism during most plant physiological processes (Jiao et al., 2023; Li et al. 2023). Additional, HXKs function as essential switches that can conduct sugar signal sensing and transduction (Granot et al. 2013). For example, in *Arabidopsis*, HXKs transport sugar signals from the chloroplast to the nucleus, activating photosynthetic genes and participating in the plastid–nucleus signaling pathway (Hsu et al. 2014). *AtHXK1* has been demonstrated to control stomatal movement, as well as coordinate photosynthesis and transpiration in *Arabidopsis*, tomato, and citrus (Lugassi et al. 2015). Mutant of *AtHXK1* exhibited reduced above-ground and root growth, diminished leaf expansion, increased apical dominance, delayed flowering and senescence, decreased growth hormone sensitivity, and heightened cytokinin sensitivity (Lugassi et al. 2015). HXKs can also regulate the triose-phosphate/phosphate translocator to influence the distribution of assimilates of chloroplasts in wheat (Sun et al. 2006). Furthermore, transgenic rice over-expressing *OsHXK5* or *OsHXK6* exhibited reduced expression of photosynthesis genes under glucose treatment, suggesting that HXKs may regulate photosynthesis in rice (Cho et al. 2009). Under elevated CO_2, photosynthesis rates in sorghum do not increase, but carbon gain is improved during drought conditions. This is attributed to improved water relations facilitated by higher photosynthetic efficiency at a reduced stomatal conductance under elevated CO_2 compared to ambient levels (Wall et al. 2001). HXKs were considered to be important in the connection between photosynthetic efficiency and elevated CO_2; however, there were rarely study about the potential function of HXKs during sugar signal sensing and transduction in sorghum.

HXKs serve as key links between sugar signaling and hormonal metabolism (Karve and Moore 2009). Given the significant role of hormonal metabolism in plant abiotic stress responses, HXKs likely play a pivotal role in facilitating plant adaptation to environmental stresses mediated by various hormones (Jiao et al., 2023; PérezDíaz et al. 2021). For instance, heightened expression of HXK in guard cells accelerates stomatal closure, primarily regulated by ABA signaling, thereby enhancing the tolerance to drought and high osmotic pressure stresses (Acevedo-Siaca et al. 2022). In *Jatropha curcas*, all *JcHXKs* were up-regulated under cold conditions, with *JcHXK3* showing significantly induction in roots (Wang et al. 2019). Similarly, in *Camellia sinensis*, the expression of *CsHXK3* was induced by cold stress and short-term ABA treatment, while *CsHXK1* was activated under salt and drought stresses (Li et al. 2017). In maize, the expressions of *ZmHXK5* and *ZmHXK6* were significantly enhanced under cold, salt, and drought stress, concomitant with the activation of ABA and GA signaling pathways (Zhang et al. 2014). These findings collectively suggest that HXKs may serve as key regulators of abiotic stress responses through their involvement in sugar metabolism in higher plants.

Generally, HXKs are encoded by a small multi-gene family in higher plants examined to date, with the first *HXK* gene isolated and identified in wheat (Saltman 1953). Subsequently, numerous *HXKs* have been identified and characterized, forming gene families in various plant species, including *Arabidopsis* (Lugassi et al. 2015), pear (Zhao et al. 2019), *Cassava* (Geng et al. 2017), cotton (Dou et al. 2022), rice (Cho et al. 2009), maize (Zhang et al. 2014), *Jatropha curcas* (Wang et al. 2019), and tea (Li et al. 2017). Sequence analysis has revealed several conserved domains, such as phosphate 1, phosphate 2, connect 1, connect 2, and adenosine phosphate-binding region, which have been utilized to identify and predict the function of *HXKs* (Yoon et al. 2021). The number of *HXKs* varies among different plant species, with some divided into different subfamilies (Karve et al. 2010). For instance, *Arabidopsis* has 6 identified *HXKs*, while maize has 9 characterized *HXKs* divided into 4 subfamilies (Karve et al. 2008; Zhang et al. 2014). In *Jatropha curcas*, only 4 *HXK* members have been isolated and characterized (Wang et al. 2019). Inrice (*Oryza sativa* L.) and pear (*Pyrus bretschneideri*), 10 *HXKs* have been identified and characterized, forming 4 and 3 subfamilies, respectively (Cho et al. 2009; Zhao et al. 2019). Additionally, 11 *HXKs* and more than 15 *HXKs* have been characterized in tea (*Camellia sinensis*) and cotton (*Gossypium hirsutum* L.) forming 6 and 4 subfamilies, respectively (Li et al. 2017; Dou et al. 2022). Furthermore, 19 *HXKs* have been found in *Brassica napus*, categorized into 4 subfamilies, indicating the potential role of *HXK* in different plant species during evolution (Wang et al. 2018). Sucrose metabolism in sorghum is linked to developmental stages, with the pathways of glucose and other hexoses extensively documented (McKinley et al. 2016). However, there is limited information available on the genome-wide identification of *HXK* gene families in sorghum, including their potential functions and their role under abiotic treatment.

In this study, we conducted homologous screening of the sorghum genome to identified 7 *SbHXKs*. Subsequently, we performed sequence and phylogenetic analyses, and identified *cis*-elements on the promoter region, elucidating their roles under abiotic stresses and developmental processes. Furthermore, the expression patterns and sub-cellular localization of *SbHXKs* were further analyzed. Collectively, our findings established a foundation for investigating the molecular role of HXKs during abiotic stress and sugar metabolism in grain sorghum.

Materials and methods

Identification of HXKs in Sorghum

Six HXK protein sequences from *Arabidopsis* genome, named AtHXK1 (NP_194642.1), AtHXK2 (AAO24584.1), AtHXK3 (NP_001321728.1), AtHXK4 (AEE32553.1), AtHXK5 (NP_188639.2), and AtHXK6 (NP_195497.1), were obtained from TAIR database (http://www.arabidopsis.org/). Ten HXK protein sequences from *Oryza sativa* genome, namely, OsHXK1 (Os07g26540.1), OsHXK2 (Os05g45590.1), OsHXK3 (Os01g71320.1), OsHXK4 (Os07g09890.1), OsHXK5 (Os05g44760.1), OsHXK6 (Os01g53930.1), OsHXK7 (Os05g09500.1), OsHXK8 (Os01g09460.1), OsHXK9 (Os01g52450.1), and OsHXK10 (Os05g31110.1), were obtained from Phytozome (https://phytozome.jgi.doe.gov). All AtHXKs and OsHXKs were used to blast sorghum protein database from Phytozome (https://phytozomenext.jgi.doe.gov/), and obtained sequences were used to verify by typical domain screening through HMMCAN (https://www.ebi.ac.uk/Tools/hmmer/search/hmmscan) (ElGebali et al. 2019). Subsequently, the incomplete sequences, redundant sequences, and transcripts were removed, and seven sequences were finally identified to name *SbHXK1* (Sobic.003G280400.1), *SbHXK2* (Sobic.003G291800.1), SbHXK3 (Sobic.009G203500.1), *SbHXK4* (Sobic.003G035500.1), *SbHXK5* (Sobic.009G069800.1), *SbHXK6* (Sobic.003G421201.1), and *SbHXK7* (Sobic.009G119100.1) (Supplemental Table 1).

Multiple sequence alignment phylogenetic analysis

Multiple sequence alignment was applied including full-length DNA sequences of 6 *AtHXKs*, 10 *OsHXKs*, and 7 *SbHXKs* by DNAman 9.0 software (Li et al. 2022). A rootless phylogenetic tree was constructed by MEGA 5 by adjacent ligation methods with the bootstrap value at 1000 repetitions. The visualization of phylogenetic tree is performed through ITOL (https://itol.embl.de) (Zhou et al. 2023b). All gaps and missing data were eliminated.

Chromosomal mapping, sequence, and conserved domain analysis

Physical analysis, chemical characterization, and conserved domains of 7 *SbHXKs* were analyzed through ExPASy (http://expasy.org/tools/). Exhibition of chromosomal locations and gene structures of all *SbHXKs* were performed via TBtools 1.0 (Chen et al. 2020). The prediction of conserved domains of all *SbHXKs*, *OsHXK*, and *AtHXK* were applied by Simple MEME Wrapper through TBtools. The protein–protein interaction (PPI) network was mapped using seven protein sequences of *Sorghum bicolor HXK* gene family in STRING 12.0 (https://cn.stringdb.org/).

Cis-elements and collinearity analysis

The 2000 bp sequence upstream of the ATG start site was obtained and considered as the putative promoter region of *SbHXK*s. *Cis*-acting regulatory elements (CAREs) were investigated through PlantCARE server (http://bioinformatics.psb.ugent.be/webtools/plantcare/html/). Various abiotic stresses and developmental responsive elements were identified and analyzed. Additionally, collinearity analysis of all *HXKs* in *Arabidopsis*, sorghum, and rice was used performed using TBtools.

Plant materials, RNA-seq, and qRT-PCR

The grain sorghum cultivar BTx623 (provided by the Biotechnology Research Center of China Three Gorges University, Yichang, China) was sown in 9 cm petri dishes. The sprouted seeds were then transferred to hydroponic boxes containing MS medium and grown for 14 days under controlled conditions (26 ℃, light intensity 1000 lx, light 16 h/night 8 h) (Cui et al. 2023). All RNA-seq samples were collected from the desired tissues labeled by specific tag, which belong to germination, seedling, reproductive, and senescence stages. Totally, two samples from germination stage were named as SEED I (24–36 h) and SEED II (48–72 h); eight samples from seedling stage were named as SHOOT (5–6 d), PRIMARYROOT (5–6 d), SEED III(5–6 d), LEAF I (7–9 d), ROOT I (7–9 d), SEED IV(7–9 d), LEAF II (10–13 d), and ROOT II (10–13 d); four samples from reproductive stage were named as ROOT III (50 d), LEAF III (50 d), INTERNODE I (50 d), and FLAGLEAF(50 d); three samples from senescence stage were named as LEAF IV (riping), ROOT IV (riping), and INTERNODE II (riping). At least three independent biological replicates were harvested in this study.

For abiotic stresses, sorghum cultivar BTx623 was grown in hydroponic boxes for 14 days. The plants were then divided as two groups: the stress group and the control group. Salt, drought, and sucrose treatment were administered using MS solution containing 200 mM NaCl, 10% PEG 6000, and 90 g/L sucrose, respectively (Li et al. 2023). The above-ground parts, including leaves and stems, of sorghum seedlings were collected for all treatment. Three biological replicates were taken for each treatment. Then, all collected samples were flash frozen in liquid nitrogen and stored at 80 °C for subsequent RNA extraction (Zhou et al. 2023a).

Total RNA extraction was performed using the Trizol method (Tiangen, Beijing, China) following the manufacturer's instructions. To eliminate DNA contamination, RNA samples were treated with DNase I. The concentration and purity of RNA were assessed using Nandrop2000 spectro photometer. The collected RNA samples were sequenced using the Illumina NovaSeq 6000 next-generation sequencing platform (Wuhan Boyuezhihe Biotechnology Co., Ltd. http://www.whbioacme.com/). The sequenced data were processed, aligned, and quantified using hisat2 and feature-counts to generate FPKM (fragments per kilobase of transcript per million mapped reads) values. Subsequently, the expression heat map was drawn using Tbtools.

First -strand cDNA synthesis was carried out using recombinant MMLV (H) reverse transcriptase (Vazyme, Nanjing, China) to prepare samples for qRT-PCR. The qRT-PCR analysis was conducted using a CFX96 Thermocycler (BIORAD, USA) with ChamQ SYBR qPCR Master Mix (Vazyme, Nanjing, China). Specific primers for expression analyses (Supplemental Table 2) were designed based on the sequences of *SbHXKs* using Primer Premier 5.0 software and synthesized by Sangon Biotech (Shanghai, China). The sorghum *Actin* gene (*SbActin*) was employed as an internal control, and the expression data were analyzed using the $2^{\Delta\Delta CT}$ methods (Chauhan et al. 2023). Statistical analysis was performed using SPSS 17.0 software.

Subcellular localization

To examine the sub-cellular localization of SbHXKs, the entire open-reading frame of *SbHXK15* and *SbHXK7* was amplified. The purified fragments of *SbHXK15* and *SbHXK7* were inserted between *35S* Promoter and green fluorescent protein (*GFP*) gene, which finally formed 6 plasmids named pLS9*SbHXK1* to pLS9*SbHXK5*, and pLS9*SbHXK7*. These six plasmids along with the empty vector *35S*::*GFP* (Control) were transformed into sorghum protoplast by PEG/$CaCl_2$ mediated transformation. The protoplast extraction method was improved based on the protocol described by He et al. (2016). The detailed procedure is as follows: Cut stems from sorghum seedlings grown under normal conditions for 1014 days into 0.51 mm fragments (approximately 1 g). Place the fragments in a 0.6 M mannitol solution and incubate them in the dark for 20 min. After removing the mannitol solution, add 10 ml of enzyme digestion solution, mix well, and incubate in the dark on a horizontal shaker at 40 rpm under vacuum for 30 min. Continue the digestion on the shaker at 40 rpm for 5 h. After digestion, increase the shaker speed to 80 rpm, and after 10 min, add an equal volume of W5 solution to the enzyme digestion solution, shaking at 80 rpm for an additional 10 min. Filter the mixture through a 200-mesh sieve into a 50 ml centrifuge tube and centrifuge at 100xg for 3 min. Discard the supernatant and re-suspend the pellet in 0.51 ml of W5 solution. Perform microscopy by mixing 10 μl of the protoplast suspension with 2 μl of FDA (fluorescein diacetate) and observe under a fluorescence inverted microscope to assess protoplast viability. Adjust the protoplast concentration to 2 × 10^5 cells/ml using MMG solution, and add 10 μg of plasmid DNA along with 110 μl of PEG-Ca solution to 100 μl of protoplasts. Mix gently and incubate in the dark at room temperature for 10 min. Then, add 440 μl of W5 solution, centrifuge at 100 xg for 3 min, and discard the supernatant. Add 500 μl of WI solution, transfer the mixture to a 24-well culture plate pre-wetted with FBS, and incubate in the dark at room temperature for 16–20 h. Finally, collect the protoplasts and observe them under a fluorescence inverted microscope. GFP expressed cell were imaged using the Fluorescence inverted microscope (Leica made in Germany, AE31 EFINV).

Results

Identification of HXK family genes in sorghum

To verify the presence of *HXKs* in sorghum, a genome-wide conserved screening was conducted based on rice and *Arabidopsis* HXK sequences using NCBICDD, pfam, and SMART databases. After removing redundant sequences, seven *SbHXK*s were identified for further analysis (Supplemental Table 1). According to their relationship with corresponding genes from rice and *Arabidopsis*, these seven *SbHXK*s were named as *SbHXK*1, *SbHXK*2, *SbHXK*3, *SbHXK*4, *SbHXK*5, *SbHXK*6, and *SbHXK*7. The genomic DNA (gDNA) and coding sequence (CDS) lengths ranged from 3668 bp (*SbHXK5*) to 6683 bp (*SbHXK5*), and from 1380 bp (*SbHXK5*) to 1767 bp (*SbHXK6*), respectively. Their molecular weight varied from 49,477.45 KDa (*SbHXK5*) to 63,497.87 KDa (*SbHXK6*). Interestingly, *SbHXK2* and *SbHXK6* exhibited similar genomic DNA length but differed significantly in CDS length, hinting at potential structural diversity among *SbHXK*s. The number of amino acids ranged from 459 to 588 aa, and theoretical pI ranged from 5.04 (*SbHXK4* and *SbHXK5*) to 6.35 (*SbHXK6*).

The stability of gene structure is a vital precondition for the conservation of gene function, and the diversity of gene structures is of great significance for the evolution of gene family. To further verify the structural conservancy of all *SbHXKs*, *HXK* protein sequences from *Arabidopsis* and rice were applied for multiple sequence alignment and motif analysis (Fig. 1). The results showed that all SbHXKs contained the conserved phosphate 1, connect 1, phosphate 2, and connect 2 domains, while substrate recognition and adenosine phosphate-binding domain were still highly conserved (Fig. 1, Supplemental Fig. 1). These results indicated that HXK gene family is highly conserved both in gene structure and functional domains in multiple plant species.

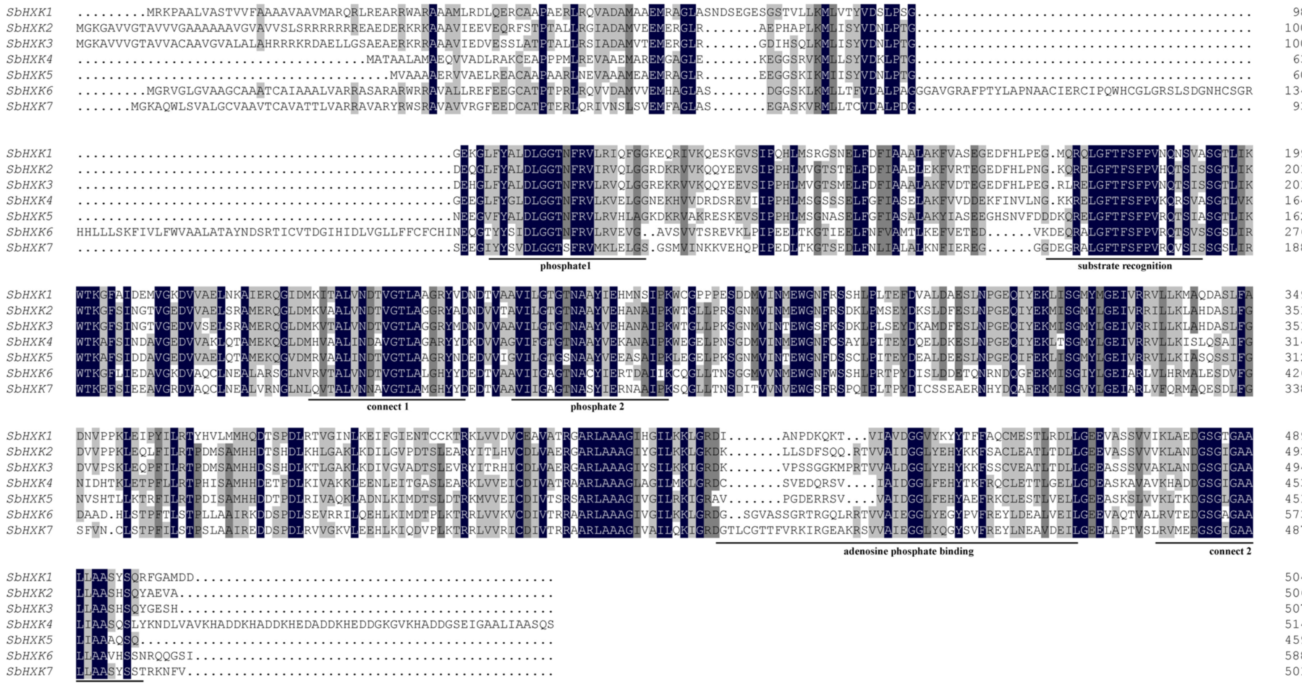

Fig. 1 Multiple alignment of HXK amino acid sequences in *Sorghum bicolor*. Identical and similar amino acid residues are highlighted in dark blue and grey shades, respectively. Core substrate recognition site, phosphate sites (I and II), connection sites (I and II), and adenosine-binding site are underlined

Phylogenetic analysis of *SbHXKs*

To investigate the phylogenetic relationships of *HXKs* between sorghum (*SbHXK17*), *Arabidopsis* (*AtHXK16*), and rice (*OsHXK110*), a rootless neighbor joining tree was constructed based on the full-length amino acid sequences (Fig. 2). According to the result, all 23 HXKs could be assigned into four distinct subgroups (I, II, III and IV). SbHXK1, SbHXK2, and SbHXK3 were clustered into subgroup I. SbHXK4 and SbHXK5 were clustered into subgroup II, while SbHXK6 and SbHXK7 were classified into subgroup IV. In subgroup I, SbHXK1, SbHXK2, and SbHXK3 exhibited high homology with OsHXK9, OsHXK6, and OsHXK5. In subgroup II, SbHXK4 and SbHXK5 showed high homology with OsHXK8 and OsHXK7, while SbHXK6 and SbHXK7 exhibit close relationship with OsHXK3 and OsHXK10 in subfamily IV, respectively. Interestingly, only AtHXK3 and OsHXK4 were clustered into subfamily III, and AtHXK6 does not cluster into any subfamilies due to this phylogenetic analysis (Fig. 2). High homology of HXKs between rice and sorghum indicated that the evolutionary divergence of the HXKs could have occurred before emergence of a common dicot and monocot ancestor.

Chromosomal location, structures, and *cis-elements* analysis of *SbHXKs*

To investigate the genomic organization and explore the evolutionary mechanisms of *SbHXKs*, chromosomal locations were further mapped. *SbHXK1*, *SbHXK2*, and *SbHXK7* were distributed on chromosome 9, and *SbHXK3*, *SbHXK4*, *SbHXK5*, and *SbHXK6* were distributed on chromosome 3 (Supplemental Fig. 2). Interestingly, all *SbHXKs* are located at both ends of chromosomes, which may have connection with their evolutionary origins. PPI network results showed that among the seven proteins, only SbHXK6 had no interaction with other proteins, and the other six proteins had certain interaction relationships (Supplemental Fig. 3).

The analysis of gene structure was performed to reveal the evolutionary relationship between *SbHXKs*. The exon–intron structure of *SbHXKs* showed that these coding sequences were interrupted by multiple introns, and the number of exons has high consistency. For example, *SbHXK6* has 12 exons, *SbHXK3* has 8 exons, *SbHXK1, SbHXK2, SbHXK4*, *SbHXK5* exon–intron and *SbHXK7* have 9 exons (Fig. 3a). The structure of exons in those *SbHXKs* were similar to each other, which indicated the conserved evolutionary mode of all *SbHXKs*. Therefore, motif analysis was further

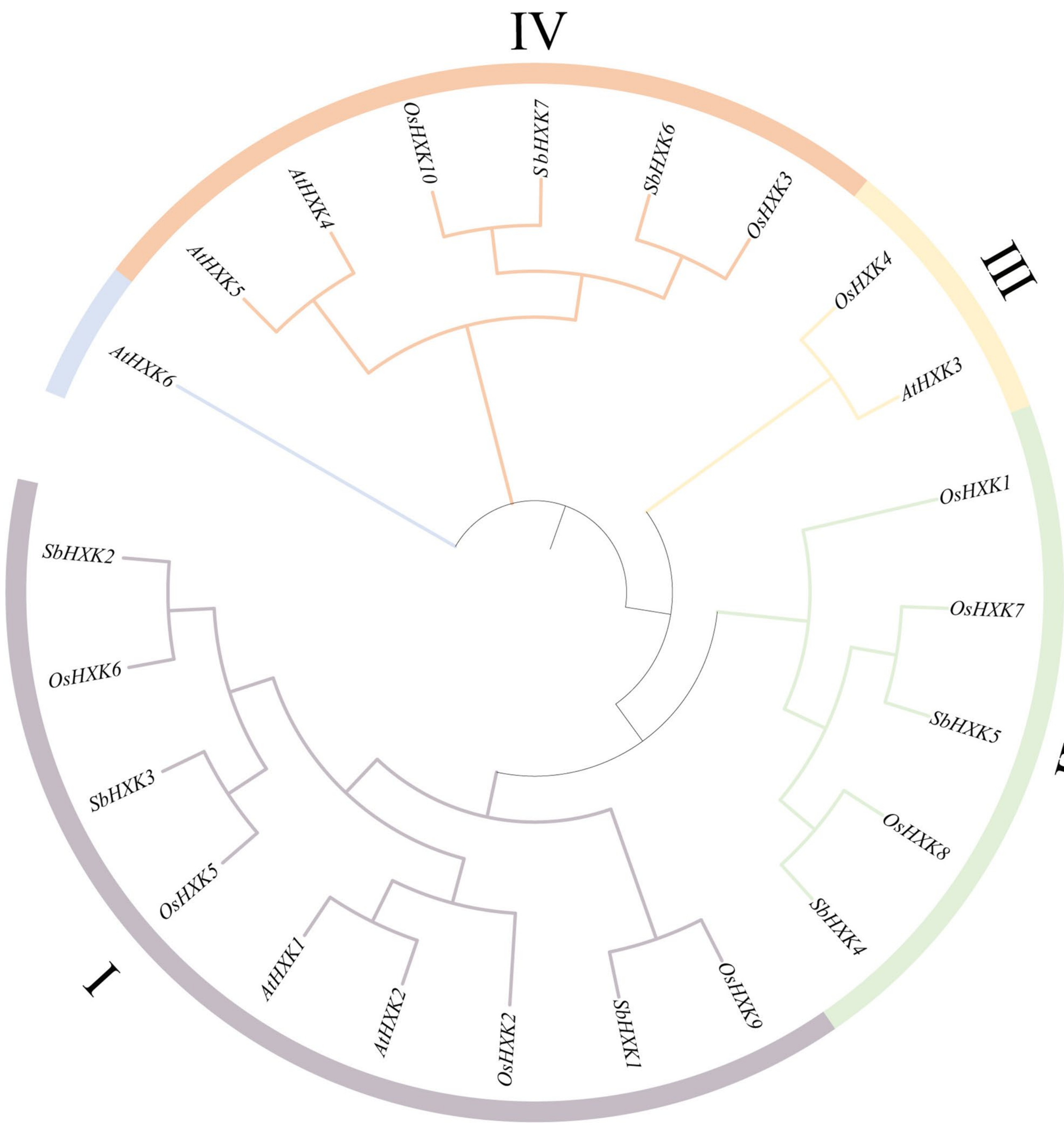

Fig. 2 Phylogeny analysis of *HXK* gene family form Sorghum and other plant species. At: *A. thaliana*, Os: *O. sativa*, Sb: *S. bicolor.* The amino acid sequences off HXK were aligned by Cluster X. Different subgroups are labeled with different colors and number (I–IV)

applied and 63 highly conserved motifs were identified in all *SbHXKs* based on the detection thresholds of motifs (Fig. 3b). All *SbHXKs* showed similar conserved motifs, which further approve the similar function of *SbHXKs* during diverse biological processes.

To explore the *cis*-elements of *SbHXKs* and its potential roles, 2000 bp upstream of the start site of *SbHXKs* were further analyzed through PlantCARE database. The results indicated that all *SbHXK* promoters contained MYC, STRE, AS1, ABRE, and MYB *cis*-elements (Fig. 3c). In higher plants, MYC, MYB, and STRE *cis*-elements are mainly corresponded to abiotic stresses, such as drought and cold stress. ABRE and AS1 *cis*-elements are widely known as hormone responsive elements, such as ABA and salicylic acid. These results suggested that all *SbHXKs* were likely to be induced by abiotic stresses and hormonal responses. In addition, there were significant differences in the number and distribution of *cis*-elements of all *SbHXKs*, which

Fig. 3 The gene structure, conserved motif organization, and cis-acting elements of *SbHXK* genes. **a** Exon–intron structures of *SbHXK*s. Exons and introns are shown as yellow boxes and black lines, respectively. **b** The conserved domains of *SbHXK*s. Different color boxes represent the different conserved motifs; **c** cis-acting elements located on promoter of *SbHXK*s. Different colors represent different type of *cis*-elements

may lead to the functional differentiation of *SbHXKs* and sugar metabolism involved in abiotic and hormone regulatory responses.

Collinearity analysis of the *SbHXKs*

To verify the evolutionary relationship of *HXKs*, a collinearity analysis was performed between *Arabidopsis*, rice and sorghum. According to this analysis, three pairs of fragments repeat genes, including *SbHXK2*/*SbHXK3*, *SbHXK4*/*SbHXK5*, and *SbHXK6*/*SbHXK7*, were identified in sorghum. And these pair of *SbHXKs* were belonged to I, II and IV subfamilies, respectively (Fig. 4). Furthermore, 2 and 3 pairs of fragments repeat genes were also identified in *Arabidopsis* and rice. However, no tandem repeat genes were founded in all *HXKs* through these 3 species revealed that fragment repeats may have important role in the evolution of *HXK* gene families. Meanwhile, 6 pair of repeated *HXKs* were identified between sorghum and rice, but there were no paired *HXKs* identified between sorghum and *Arabidopsis* (Fig. 4). Altogether, these results indicated a more closed relationship between sorghum and rice, both are monocots, than that between sorghum and *Arabidopsis*. Therefore, we speculated that *HXK* gene families might originate from a common ancestor.

Expression patterns of *SbHXKs* in different physiological processes

To determine the expression patterns of *SbHXKs* in different tissues at different developmental stages, we utilized an existing RNA-seq database from our laboratory (unpublished) to display the expression levels of all *SbHXKs* at different fertility stages (Supplemental Fig. 4). The heatmap result showed a significant different expression profile of all *SbHXKs*. The expressions of different *SbHXK*s showed different patterns. For example, *SbHXK1* and *SbHXK3* were grouped into one cluster. However, *SbHXK1* was highly expressed in root and mature leaf, while *SbHXK3* was highly expressed in young root and reached the highest level at 79 d after germination (Supplemental Fig. 4). *SbHXK2* showed highly expression in seed at 56 d after germination, which indicated the potential role during early leaf development. *SbHXK4* also showed high expression in seed germination and was cluster with *SbHXK2*. *SbHXK5* and *SbHXK6* were grouped as

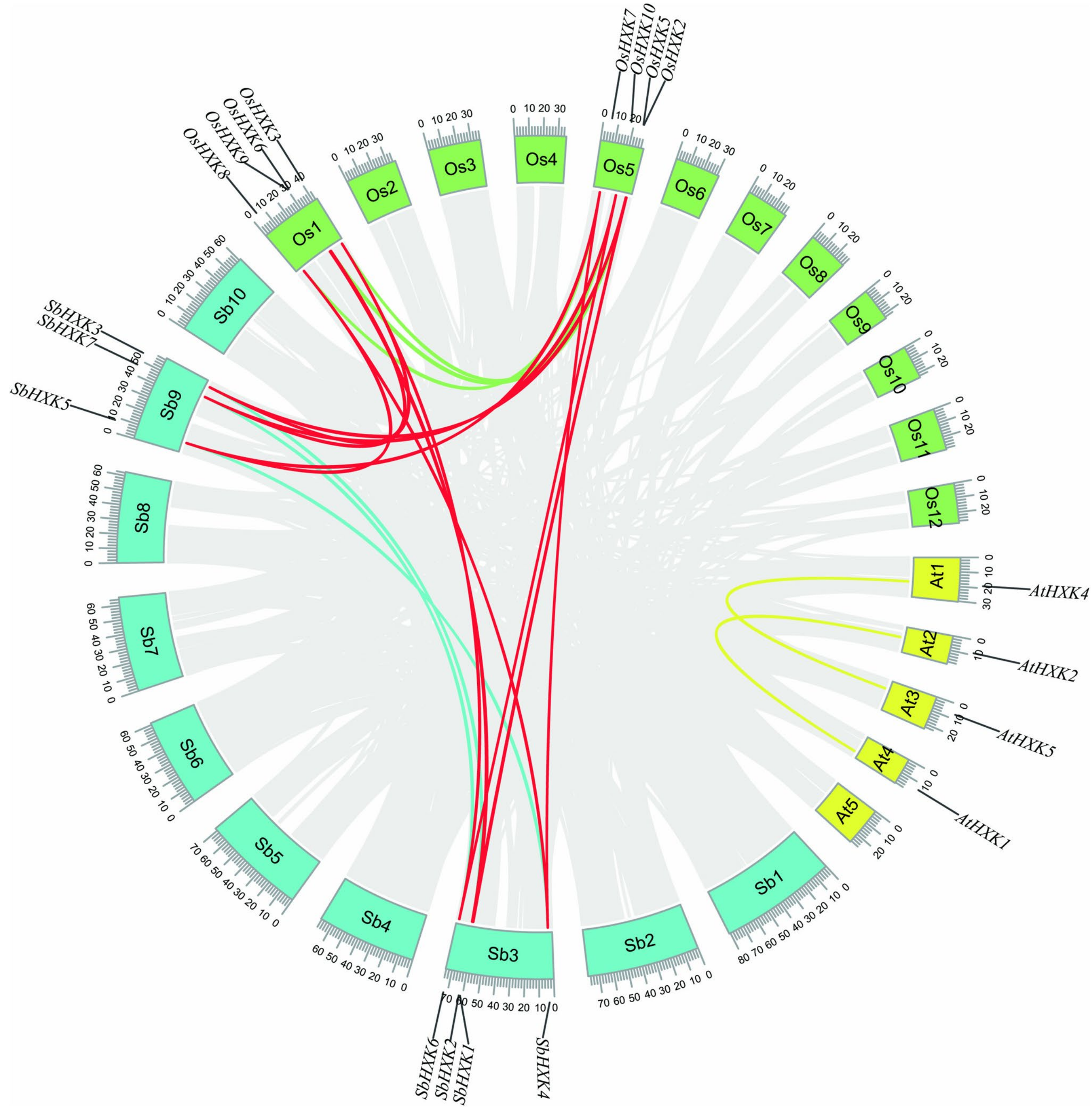

Fig. 4 Assessment of the *HXK* gene duplication between Sorghum, *Arabidopsis*, and rice. Different chromosomes of sorghum, *Arabidopsis,* and rice are shown as blue, green**,** and yellow boxes, respectively. Different HXKs from different plant species are labeled. Duplication links between different species are represented by red lines, and duplication links in the same specie are labeled by blue, green**,** and yellow lines, respectively

one cluster, and they were both highly expressed in young tissues including seed, shoot, and primary root. *SbHXK5* was highly expressed in leaf tissues, and *SbHXK6* was especially expressed in root tissues, respectively. Moreover, *SbHXK7* was only highly expressed in internode (50d), which may indicate the potential function of *SbHXK7* during material transportation. These results suggest that *SbHXKs* may play specific role through different physiological processes in sorghum.

To better elucidate the physiological functions of specific *SbHXK* during biotic stresses, qRT-PCR was applied to further investigate the expression profiles of *SbHXKs* under salt, drought, and sucrose stresses (Fig. 5). In the plants under salt simulation, *SbHXK1*, *SbHXK3*, *SbHXK4*, *and SbHXK5* were

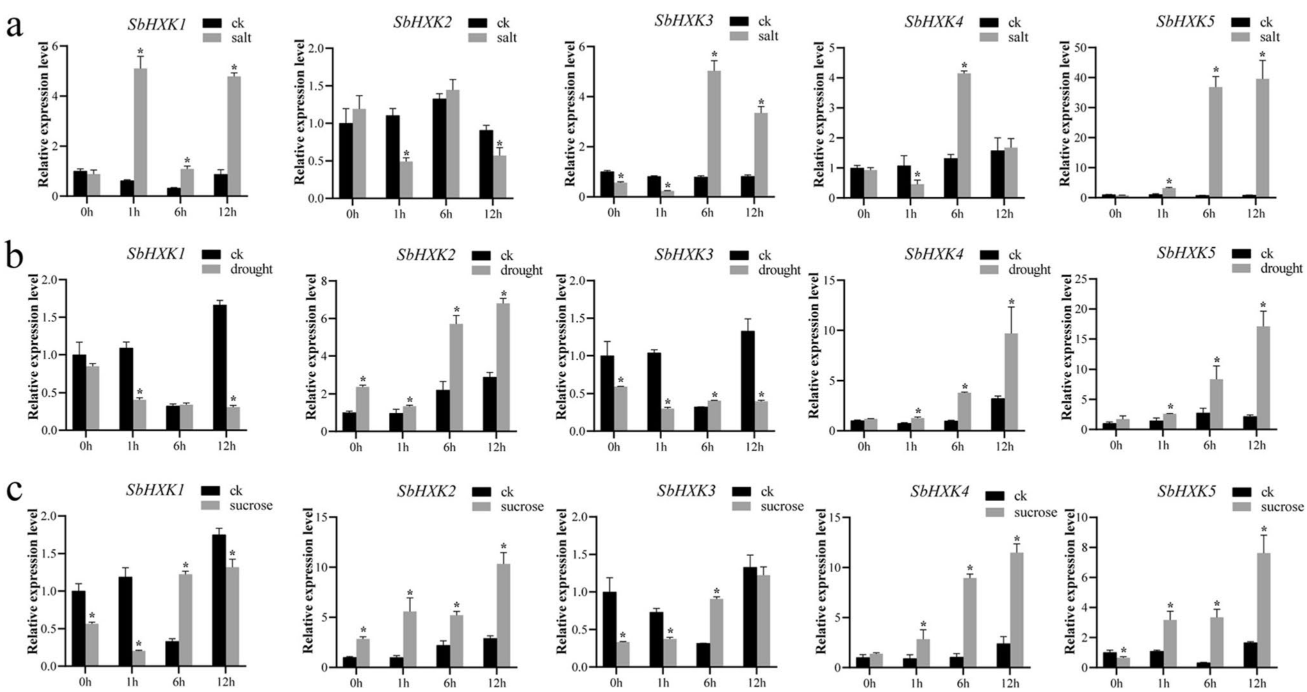

Fig. 5 The expression patterns of *SbHXKs* under different treatments. **a** Expression of 5 HXK genes under drought stress. **b** Expression of 5 HXK genes under salt stress. **c** Expression of 5 HXK genes under sucrose treatment. Expression of *SbHXKs* are compared between control and different treatments, and labeled as black and gray bars, respectively. Data represent mean ± standard deviation (SD) with three biological replicates

induced significantly at 1 h or 6 h after salt treatment, and *SbHXK2* was suppressed at 1 h after salt stress (Fig. 5a). Similarly, when sorghum seedlings were treated under 10% PEG 6000, *SbHXK2*, *SbHXK4*, and *SbHXK5* were significantly activated at 6 h under drought stress, while *SbHXK1* and *SbHXK3* were significantly suppressed, respectively (Fig. 5b). Moreover, *SbHXK2*, *SbHXK4*, and *SbHXK5* were induced significantly respond to sucrose treatment, while *SbHXK1* and *SbHXK3* showed no obvious change under sucrose treatment (Fig. 5c). Altogether, *SbHXK2*, *SbHXK4*, and *SbHXK5* may play very important functions particular under high osmotic pressure.

Subcellular localization of *SbHXKs*

Previous studies had indicated that most HXK family members in *Arabidopsis* are associated with mitochondrial, and the Nterminal trans-membrane peptide that target to mitochondria supports this evidence (Karve et al. 2008). However, most HXK family members in sorghum and rice did not contain this Nterminal trans-membrane peptide, may be indicating the different sub-cellular localizations (Cho et al. 2009). To further verify this hypothesis, full length of *SbHXKs* were cloned into *GFP*fused vectors. Constructed vectors were then transformed into sorghum protoplast for microscope visualization. The results showed that the SbHXK15 and SbHXK7 fusion proteins with green fluorescent signal were localized in cytoplasm, which supported this hypothesis with different localization (Fig. 6). Since most hexose phosphorylation, energy production, and other physiological processes were regulated in cytoplasm, our results may supply a new sight that HXK proteins may play different roles during hexoses phosphorylation between monocot and dicot plants.

Discussion

Hexose sugars are vital as the primary carbon source for energy storage, significantly contributing to cellular life and metabolism (Morgan and Joshua 2023). For their utilization, hexoses must be phosphorylated by hexokinases (HXKs), which play diverse roles in phosphorylating fructose and glucose (Li et al. 2017). HXKs have been found across many plant species. So far, knowledge about the function of the HXK gene in sorghum remains scarce. With the development of sequencing technology and the completion of whole-genome sequencing, the identification and characterization of *HXK* gene family has been carried out in many plant species (Karve et al. 2010). Preliminary identification of *HXKs* in rice and *Arabidopsis* indicated that *HXK* family was consist of 6–10 members that can act as glucose sensor

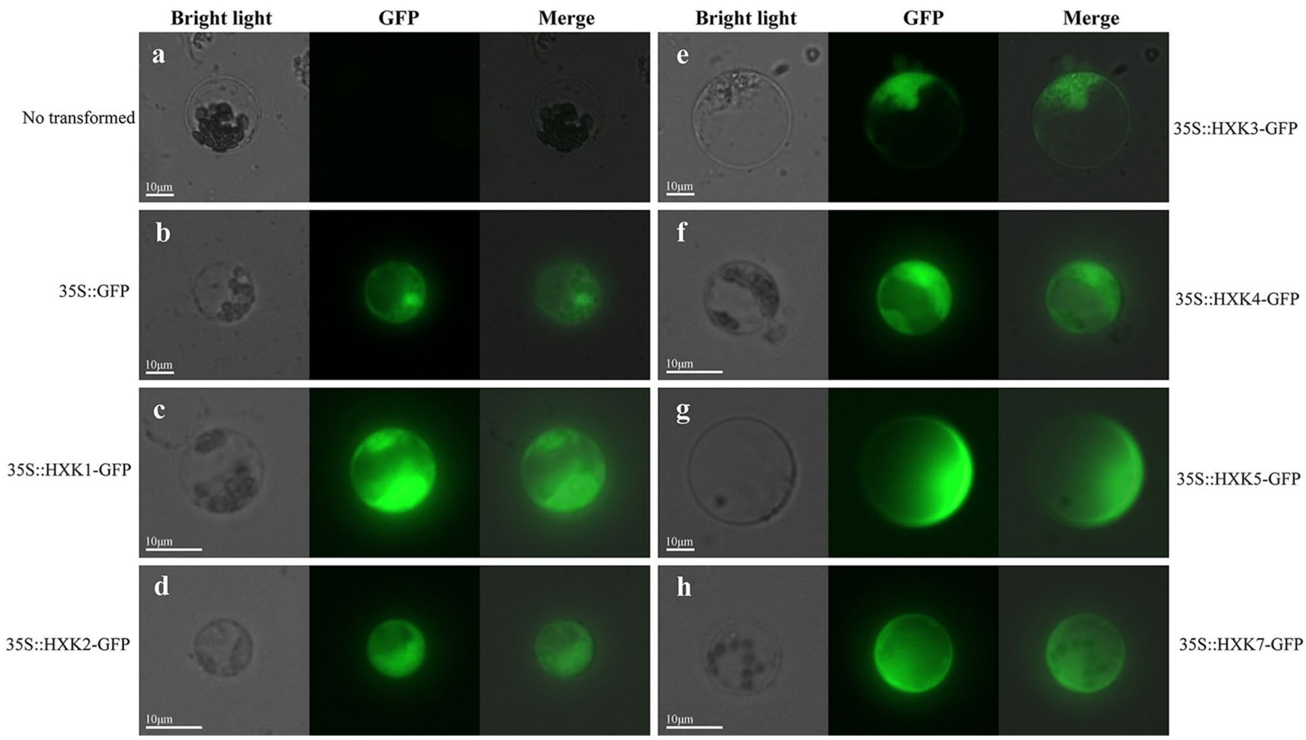

Fig. 6 Subcellular distribution of *HXK* protein in sorghum protoplasts. **a** No transformed. **b** Negative control with 35S::GFP vector. **c** Expression of SbHXK1 in protoplast. **d** Expression of SbHXK2 in protoplast. **e** Expression of SbHXK3 in protoplast. **f** Expression of SbHXK4 in protoplast. **g** Expression of SbHXK5 in protoplast. **h** Expression of SbHXK7 in protoplast. The scale bar is 10 μm

or transduction molecules (Halford et al. 1999). In this study, 7 *SbHXKs* were identified in sorghum and highly conserved with 6 *AtHXKs* and 10 *OsHXKs* (Fig. 2). By comparing with *SbHXK*s with HXKs in maize, the lengths of *HXK*s in both C4 crops are quite similar, but 9 ZmHXKs are distributed on 3 chromosomes, while 7 SbHXKs are distributed on 2 chromosomes (Zhang et al. 2014) We also found that all 7 *SbHXKs* have conserved phosphate 1, connect 1, phosphate 2, and connect 2 domains, through the homologous sequence alignment between sorghum, *Arabidopsis,* and rice (Fig. 1, Supplemental Fig. 1). Moreover, phylogeny analysis and collinearity analysis indicated that *SbHXKs* are more closed to *OsHXKs*. For example, *SbHXK2* and *OsHXK6* were clustered in subgroup II, and also exhibit the close collinearity. *SbHXK6* and *OsHXK3* were also clustered in subgroup IV, and show the close evolutionary (Figs. 2, 4). There were no paired *HXKs* between sorghum and *Arabidopsis* (Fig. 4). Interestingly, all *HXKs* were: These findings suggest that HXKs exhibit a close evolutionary relationship across various plant species, particularly between rice and sorghum, indicating that HXKs may have evolved from a common ancestor.

Sugar signaling plays a crucial role in linking sugar production with plant development and responses to environmental cues. In C3 plants, it is well recognized that excessive sugar accumulation in source leaves can negatively affect photosynthesis and productivity (Chen et al. 2024). However, our grasp of the molecular mechanisms behind these feedback regulations, especially in C4 plants, is still quite limited. HXK is known for its role in sugar feedback regulation of photosynthesis in the C3 dicot *Arabidopsis* and the C3 monocot rice, and to a lesser extent, in the C4 monocot maize (Chen et al. 2024). Previous studies revealed that HXK proteins contain conserved domains, including phosphate 1, phosphate 2, connect 1, connect 2, sugar binding, and adenosine-binding site, which are important to plant HXKs and essential for their enzymatic functions (Yoon et al. 2021). In our study, multiple sequence alignments of sorghum HXK proteins highlighted highly conservation that are consistent with the reported HXKs, such as phosphate 1, phosphate 2, adenosine, and sugar-binding (Fig. 1; Supplemental Fig. 1). By comparing the HXK sequence of sorghum, rice, and *Arabidopsis*, we found that the functional domain sequences in sorghum are more similar to those in rice, while they exhibit greater discrepancies with *Arabidopsis* (Supplemental Fig. 1). *SbHXK6* exhibited some observable deviations, and there is a 92-amino acid inserted ahead of phosphate 1 domain. Furthermore, *HXK*s do not show preferential expression between plant source tissues and sink tissues, which may related to sugar sensing and signal transduction (Cho et al. 2009). In our study, *SbHXK5* and *SbHXK6* are considered as main glucose sensor in monocot plant, which were highly expressed in source tissues. *SbHXK3* and *SbXHK7* exhibited high expression levels in root and internode tissues, both of which are classified as source tissues. In contrast, SbHXK1, SbHXK2, and SbHXK4 are predominantly expressed in seed and leaf tissues, which are classified as sink tissues (Supplemental Fig. 4). Therefore, we propose that *SbHXK*s show preferential expression in source and sink tissues, reflecting the differences in sugar sensing and signaling between C4 and C3 plants.

Hexokinases are crucial enzymes in plant growth and development (Cho et al. 2009; Granot et al. 2013; Li et al. 2023). For example, *OsHXK10* in rice plays a role in regulating plant reproduction (Xu et al. 2008). In *Arabidopsis*, *AtHXK1* signaling and catalytic activity are crucial for plant development, and *AtHKL3* is exclusively expressed in flowers (Morgan and Joshua 2023). *OsHXK10* only expressed in rice flowers, and *OsHXK3* and *OsHXK7* were abundantly expressed in seed coats, while *OsHXK2*, *OsHXK4*, *OsHXK5*, *OsHXK6*, and *OsHXK8* were abundantly expressed in endosperm (Cho et al. 2009). In maize, *ZmHXK5* is highly expressed in tassels, while *ZmHXK6* is expressed in leaves and tassels (Zhang et al. 2014). In our study, *SbHXK1* is highly expressed in primary root, and *SbHXK3* is highly expressed in most young tissues. Expression of *SbHXK7* was only detected in internode at 50 days after germination, and *SbHXK2* is highly expressed in seed at 56 days after germination, respectively (Supplemental Fig. 4). Sorghum is a crop known for its adaptability thrive under various extreme conditions (Taylor et al. 2014). While many studies have highlighted the significant role of the *HXK*s in plant stress responses, research on the involvement of *SbHXK*s in sorghum's stress responses remains unclear. ABA is a key plant hormone whose concentration within the plant is close influenced by external environmental stress factors. Plants respond to environmental stress through interaction of transcription factors with a handful of cis-regulatory elements (Ankur and Aryadeep 2023). In our study, we identified the cis-elements of each *SbHXK*, finding that the promoter region of all *SbHXKs* contains ABRE cis-elements, which are known to be associated with ABA signaling and play a crucial role in regulating stomatal conductance during abiotic stress (Fig. 3b). MYB transcription factors are common in plants and engage with MYB cis-elements, which help regulate responses to different stress conditions (Baltasar et al. 2023). Additionally, we found that all *SbHXK*s possess various MYB *cis*-elements within their promoter regions. Besides, other *cis*-elements, such as MYC, STRE, and AS-1, were identified as being associated with responses to abiotic stresses (Fig. 3b). Additionally, the expression of *SbHXK*s was analyzed using qRT-PCR under various stress conditions, revealing that *SbHXK1*, *SbHXK3*, *SbHXK4*, and *SbHXK5* are induced by salt treatment (Fig. 5a). With drought and sucrose treatment, *SbHXK2*, *SbHXK4*, and *SbHXK5* are induced, while *SbHXK1* and *SbHXK3* are suppressed significantly (Fig. 5b). These findings indicate that *SbHXK*s may have a significant role in sorghum's developmental stages and stress responses, providing an important insight for investigating the mechanisms behind its resilience.

Sorghum is noted for its high biomass and efficient photosynthetic capabilities, making it a promising candidate for agricultural and biofuel production. Our study demonstrates that *SbHXK*s are crucial for plant development, stress responses, and sugar sensing. Understanding their roles could enable genetic engineering approaches to enhance grain size, nutritional content, and overall yield in sorghum. Furthermore, *SbHXK*s are important in carbohydrate metabolism, which is essential for maximizing biomass accumulation and energy conversion processes. By optimizing HXK activity, we can enhance the efficiency of biomass production, critical for increasing biofuel yields from plant materials. However, due to the difficulties in sorghum genetic transformation, we were unable to obtain transgenic plants expressing *SbHXK*s, thus preventing us from verifying the functions of *SbHXK*s in plant development, stress resistance, and sugar sensing. Additionally, the lack of biochemical analysis hindered our ability to further analyze the functional differences between *SbHXK*s and those in C3 plants. These limitations collectively constrained our ability to validate the bioinformatics results. In summary, *SbHXK*s are integral to shaping agricultural practices and bioenergy strategies, with the potential to boost resilience and improve the efficiency of biofuel production in sorghum.

Conclusion

In this study, based on the whole-genome analysis of sorghum, seven *SbHXKs* were identified, and the sequence, structure, chromosome localization, phylogeny, collinearity, and cis-acting elements were analyzed, and seven *SbHXKs* were separated into three clusters. Expression profiles of *SbHXKs* in different tissues indicated the preferential expression between source and sink tissues, reflecting the differences in sugar sensing and signaling between C4 and C3 plants. In response to abiotic stresses, the expression of *SbHXK*s reflecting their potential functional differences. The current work further provides comprehensive information about the *SbHXK* gene family, which will facilitate future cloning and analysis of *SbHXK* gene function in the sorghum.

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Data availability The data that support the findings of this study are available on request from the corresponding author.

Declarations

Conflict of interests The authors declare that they have no competing interests.

Ethical approval Not applicable.

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